PA Consistency Machines: Comparison and Further Reduction Experiments

Technical research report

14 November 2026 (UTC)

Abstract

We inspect all 27 top-level archives in the requested incoming directory, including their nested archives, and compare them with the imported PA196 and PA209 projects and the extensive PA196–U208 comparison tree. The census finds 2,447 occurrences of complete numeric binary transition tables, representing 531 byte strings and 522 distinct labeled transition functions. These figures include historical copies, rejected experiments and controls; they are not counts of established consistency machines. The 14 occurrences of 195-state tables belong to six rejected diagnostic constructions. We give a precise logical audit, a proof-preserving family of independently reparameterized axiom heads, new finite searches that score physical decoder sharing, and a correction to the interpretation of intermediate dispatcher returns. **No machine with fewer than 196 working states and the requested PA-provable consistency biconditional is established by this investigation.** The archive is a reproducible research result and comparison, not a claimed solution to that remaining task.

1 Target, conventions, and result status

The requested result is a completely specified deterministic, single-tape, two-symbol Turing machine T, started on an all-zero two-way infinite tape, such that

$$|Q_T| \le 195, \qquad \mathsf{PA} \vdash \big(\neg\,\mathrm{Halt}(T, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})\big). \tag{1}$$

Here Q_T counts working controls; a separate halting control is excluded. Every working control has one transition for each symbol in $\{0, 1\}$, and every transition moves left or right. We retain the binary Busy Beaver convention of the supplied constructions.

The source snapshot is the Git commit

`906367250a7f96c923147b10df392294bced7be5`

of VladimirReshetnikov/ProveIt. All repository statements in this report refer to this immutable revision. In particular, the state of a proof in this snapshot takes precedence over informal descriptions in earlier conversations or archive titles.

There are two independent completion conditions in (1). A smaller source expression is not a smaller physical machine. A physical state reduction does not itself prove the consistency calibration. This report keeps four levels of evidence distinct:

1. a syntactically complete table and an exact working-state count;
2. preservation of a particular source program or command calculus;
3. exact blank startup and arbitrary-value primitive simulation;
4. both consistency implications formalizable inside PA.

A symbolic executable checker is stronger than testing finitely many numeric register values when its loop rules are justified, but is still a different artifact from a kernel-checked proof or an exported theorem of PA.

1.1 What is delivered

The accompanying archive contains this article in LaTeX and PDF, the complete archive inventory with hashes and member paths, a per-archive comparison, new search programs and their actual receipts, selected pinned source inputs, and proof-audit and scanner-witness artifacts. Large historical archives are identified by hash rather than duplicated. Reproducing the complete census requires the pinned ProveIt checkout; reproducing the selected local searches uses the included compact inputs.

The negative outcome is specific: the required machine was not constructed. None of the searches proves that 196 states is a lower bound. Rejected simulation certificates also do not, by themselves, prove that every associated blank-run consistency statement is false.

2 Complete corpus comparison

2.1 Inventory and deduplication

The inventory visits all 27 ZIP files in `docs/incoming`, recursively opens nested ZIPs, and inspects repository tables under `Computability/BusyBeaver/Research/PA`. A numeric table is counted when its rows provide exactly two transitions for every working label. Duplicate state-symbol rows are rejected. Counts are:

Quantity	Count
Top-level incoming archives	27
Complete numeric table occurrences	2,447
Distinct raw byte strings of such tables	531
Distinct labeled transition functions	522
Occurrences with exactly 195 working states	14
Distinct raw byte strings among those occurrences	8
Distinct rejected 195-state diagnostic constructions	6

The distinction between a byte string and a transition function accounts for comments, headers, whitespace and row ordering. Native/dense presentations of two of the 195-state constructions were independently matched using their documented starts, native 74 and dense 0. The remaining diagnostic constructions are kept separate.

The census recognizes complete numeric tables in `.tm`/`.txt` archive members and repository `.tm` files. Arbitrary JSON-encoded tables and generators are not counted as literal tables; their main design reports are nevertheless included in the comparison. It would be misleading to label the 522 transition functions as 522 verified PA machines.

Some native tables lack an explicit start directive. The inventory retains a *provisional* rooted-graph index for navigation, with an explicit `start_inferred` flag. Such a provisional index is not used as an exact whole-corpus count of inequivalent blank machines. Start conventions must be resolved from their accompanying documentation first.

2.2 The principal design families

Family	States	Distinguishing result and qualification
Imported PA209	209	3,948 source sites; direct predecessor of PA196. Its exposed internal consistency implication is one-way.
Primary U208	208	3,909 source primitives in the reviewed lineage; a local history fold with exact startup and symbolic instruction contracts. The original nine presentations reduce to two principal graph classes.
Extended U208	208	Eighteen additional helper variants in the larger research packet, giving twenty reported same-size graph classes with startup and primitive-contract checks in that U208 lineage.
Imported PA196	196	3,812 source sites; 159 distinct internal source nodes; changed quantifier coding, axiom emitter and physical contextual sharing.
Rearranged PA196	196	Several different exact-CFG layouts, shorter paths, register renamings, and command variants. State savings from these presentations cannot automatically be added.
Diagonal-vacuity	196	191 AX macros and 3,700 sites; an earlier 199-state realization is improved to 196 in a later packet.
Separate diagonal	196	190 AX macros and 3,673 sites; independently checked same-size realization and a scanner-oriented 196-state layout.
NEG plus MP	197	Shorter arithmetic negation plus successful MP returning in H: 3,573 source sites. Shorter source does not yield a smaller table.
Rejected diagnostics	195	Six constructions have failures or missing obligations in their proposed verification routes. None supplies (1).

The main source for the imported comparison and the distinction between PA209 and U208's different U209 parent is the repository comparison. The PA196 paper explicitly decomposes its 196 controls into 36 engine controls, 13 leaf-routing controls and 147 source-reader controls. The six pairs of complementary AX nodes already folded in that machine are not six further available savings.

2.3 Why the U208 deletion does not immediately transfer

In the selected native U209 parent, the U208 construction uses a state entered from a predecessor whose two branches have opposite movement directions:

$$163, 0 \mapsto (0, L, 211), \qquad 163, 1 \mapsto (1, R, 211).$$

The target and a keeper have a matching branch, while the incoming history selects the appropriate behavior on the other branch. A local tape simulation establishes the relevant one-, three- and seven-step blocks.

A direct scan of PA196's 392 rows finds no working control with that exact mixed-entry pattern to one target. Therefore renaming labels cannot transplant this particular literal motif. This is only a statement about that motif; different helpers, more predecessor history, different source layouts or different representations remain possible. The U208 fold's correctness also does not identify its parent with James Chang's distinct 209-state machine.

2.4 Per-archive comparison

Appendix A records every top-level incoming archive. The JSON catalogue additionally gives hashes, exact member paths and all parsed table occurrences. Nested old machines explain the large occurrence count; they do not represent independent new improvements.

3 The logical endpoint: what is proved and what remains

Let H_{196} be the arithmetized halting sentence for the exact imported table and let $N_{196} = \neg H_{196}$. The shipped declaration `PA196.Arithmetic.pa_proves_nonhalting_consistency` has the form

$$\mathsf{PA} \vdash N_{196} \to \mathrm{Con}(\mathsf{PA}). \tag{2}$$

The same project proves literal nonhalting in the Lean metatheory and proves two-sided PA independence. These closed corollaries use Foundation's standard-model consistency and soundness facts in Lean's metatheory. Neither assertion supplies

$$\mathsf{PA} \vdash \mathrm{Con}(\mathsf{PA}) \to N_{196}. \tag{3}$$

This limitation is already explicitly recognized in the imported-snapshot README, `COMPARISON.md`, and PA-BRIDGE.md. It is not a new defect discovered by this investigation, and it is not a claim that (3) is false or unprovable.

3.1 A precise sufficient route for the missing direction

Write $\mathrm{Tr}_T(t)$ for the decidable predicate that t codes a finite blank-to-halt trace of T. Write $\mathrm{NCert}(d, 1)$ for acceptance of a native derivation of the distinguished false formula, and $\mathrm{Prf}_{\mathsf{PA}}(p, \bot)$ for the chosen PA refutation predicate. A sufficient route is to give primitive-recursive proof transformations e, f, with PA proofs of

$$\mathrm{Tr}_T(t) \longrightarrow \mathrm{NCert}(e(t), 1), \tag{4}$$

$$\mathrm{NCert}(d, 1) \longrightarrow \mathrm{Prf}_{\mathsf{PA}}(f(d), \bot). \tag{5}$$

Existential quantification yields $H_T \to \neg\,\mathrm{Con}(\mathsf{PA})$, and classical contraposition yields $\mathrm{Con}(\mathsf{PA}) \to \neg H_T$. Together with the appropriate forward implication, this proves the requested biconditional.

Equation (4) must handle arbitrary internally coded finite traces, including nonstandard lengths in nonstandard models of PA. It needs startup, determinism, positive primitive durations, totality of nonhalting segments, and extraction through a possibly unfinished last segment. A source-to-tape simulation in the other direction does not automatically give this reverse extraction.

Equation (5) is a uniform proof transformation, not a truth predicate for all arithmetic formulas. Under the intended bit-membership interpretation, native axiom instances require uniform PA proof templates. The separation template invokes PA induction with the translated input formula. All coding bounds, variable conventions, freshness conditions and stack certificate interfaces must be attached to the exact native calculus.

There is a particular interface issue in translating quantifier encodings: PA196 uses the property code as a bound-variable label. Reversing the arguments of a pairing function changes a property's formula code but does not automatically change that label. A correct transport therefore needs a generic fresh-binder theorem or a verified renaming, not just a statement that the pairing arguments are swapped. The existing repository notes identify this obligation explicitly.

4 A sound enlarged family of axiom emitters

The physical search need not be restricted to reassociating one fixed source word. It may also vary a presentation while preserving the exact family of available axiom instances. We record the justification because this is an appropriate way to broaden the search without weakening its logical target.

Use the imported numerical constructors $\mathsf{I}, \mathsf{M}, \mathsf{U}, \mathsf{E}$, with the transposed quantifier coding $\mathsf{U}(d, h) = 4\,\mathrm{pair}(h, d) + 2$.

Let $\mathsf{N}(B) = \mathsf{I}(B, 1)$ and $\mathsf{Ex}(d, B) = \mathsf{N}(\mathsf{U}(d, \mathsf{N}(B)))$. In emission order the six heads are

$$\begin{aligned}
h_1 &= \mathsf{I}(\mathsf{Ex}(D, \mathsf{U}(D, H)), H),\\
h_2 &= \mathsf{I}(\mathsf{U}(D, \mathsf{U}(A, H)), \mathsf{U}(A, \mathsf{U}(D, H))),\\
h_3 &= \mathsf{I}(\mathsf{M}(H, D), \mathsf{I}(\mathsf{E}(H, A), \mathsf{M}(A, D))),\\
h_4 &= \mathsf{I}(\mathsf{E}(H, D), \mathsf{I}(\mathsf{M}(A, H), \mathsf{M}(A, D))),\\
h_5 &= \mathsf{Ex}(D, \mathsf{E}(H, D)),\\
h_6 &= \mathsf{I}(\mathsf{U}(D, A), \mathsf{I}(\mathsf{U}(D, \mathsf{I}(A, H)), \mathsf{U}(D, H))).
\end{aligned}$$

The command prefix that saves the old top, produces a generalization and produces the restricted propositional head ρ_8 is kept unchanged.

Theorem 1 (Independent head reparameterization)**.** *For each $i \in \{1, \ldots, 6\}$ choose an arbitrary permutation π_i of the three coordinates (D, H, A). Replace $h_i(D, H, A)$ by $h_i(\pi_i(D, H, A))$, and emit the resulting six heads in any order. With the same gate family, generalization rule, propositional head and modus ponens, the native derivability relation is unchanged. The induced proof translations in both directions are primitive recursive.*

Proof. Each π_i is a bijection of $\mathbb{N}^3$. Therefore

$$\{h_i(\pi_i(v)) : v \in \mathbb{N}^3\} = \{h_i(v) : v \in \mathbb{N}^3\}$$

as a set of numerical formula codes. This also applies when a head ignores one coordinate. At a proof leaf using $h_i(v)$, supply the parameters $\pi_i^{-1}(v)$ in the new presentation. Leave other axiom leaves unchanged and recurse through modus ponens and generalization. The inverse transformation uses π_i. Finite-tree recursion and finite coordinate permutations are primitive recursive. No consistency or semantic soundness assumption is used. □

At the command interface, the existing DISCARD operation sets $d = h = 0$. Consequently $\mathrm{AX}(c, x, y)$ realizes $D = x$, $H = y$, $A = c$, so every parameter triple is available. The compiler chooses the inverse-permuted parameters and adjusts the fixed Shift selector to the head's new stack position. The six heads still have code 0 or a code at least 2: implication codes are multiples of 4, and the existential head is itself an implication code. This preserves the selector's required avoidance of false code 1. In this original 201-token emitter family there are still nine pushed values including the saved top. The separate-diagonal-head 190-macro emitter used elsewhere in the experiments has one additional axiom head, ten pushed values, and correspondingly adjusted selectors. None of this claims that an arbitrary new physical allocation is correct.

4.1 Additional justified variants and a rejected shortcut

The executable emitter reproduces the exact 201-token baseline from `PA196/Source/AXProgram.lean`. It also exposes six independent bit choices: exchange curried antecedents in h_3, h_4, h_6; exchange equality and membership vacuity in the outer gate; exchange the two conjunction terms in separation; and exchange the first two antecedents in the relevant separation implication. The head changes use implicational exchange. The gate changes use propositional equivalence and the retained quantifier-distribution/generalization machinery. The original h_1 instance family remains present.

Swapping D, H throughout the gate alone is also an exact reparameterization. The property code A stays fixed, and the guard $A \neq D, H$ is symmetric. We do not allow an unproved swap of A with an object-label register: A is also the property's formula code and the designated binder.

An initially considered replacement of h_1 by a negation-based contrapositive is deliberately disabled. The native calculus derives explosion using h_1 itself. Appealing to explosion to recover h_1 from its proposed replacement would be circular. This rejected option is absent from the completed searches.

5 Fresh reduction experiments

This section reports experiments performed during the present investigation, not just results copied from the incoming packets. Exact counts and input identities are retained in the accompanying JSON receipts.

5.1 Global recompression

We tried 20,000 randomized repeated-pair recompressions of the closed AX sequence, also supplying externally used subexpressions to encourage sharing. The unchanged full program has 159 distinct internal source nodes. The best newly sampled source has 162, with no improvement. This experiment is a heuristic over associative grammars; its failure is not an optimality proof. It also illustrates why a grammar search must be scored against the physical allocation rather than an informal count of shortened arithmetic expressions.

5.2 Exhaustive direct head substitutions

For each of the two supplied 196-state diagonal variants, we independently permuted (D, H, A) in each of the six heads, retaining the binary association. This gives $6^6 = 46,656$ cases per grammar and 93,312 cases in total. The imported physical template contributes 134 fixed controls, so a successful sub-196 allocation would need an AX contribution of at most 61.

For these two fixed associations, the identity parameter assignment is the unique case with matching bound 62. Every nonidentity assignment has bound at least 66. The eligible cases also receive the exact simultaneous-helper check. Thus this particular exhaustive family gives no improvement. The statement is restricted to the retained associations and imported pair-fold template; reassociation, different helpers or another engine are not excluded.

All $6! = 720$ direct reorderings of the six heads were additionally examined for each of the two grammars, 1,440 cases. They likewise do not reduce the physical contribution below 62. Further mixed semantic/association searches broaden this family, but remain heuristic. Their completed proposal counts are recorded in the generated experiment summary, rather than inferred from nominal budgets.

5.3 Longer exact decoder helpers

The incoming compiler's standard context fold distinguishes complementary reader nodes by looking at the immediately preceding path bit. Its familiar helper trace consists of a left step, a right step, and a right step. We extended the search to arbitrary deterministic computations in a fixed finite window, allowing the first write to flip the current bit provided every touched bit is eventually restored. The replacement conflicting row may write either bit, moves left, and enters an existing state; subsequent helper steps use retained rows.

For a width w, a local configuration comprises a working control, a head position in $\{-w, \dots, 0\}$, and a binary word of length $w + 1$. There are at most

$$|Q|(w + 1)2^{w+1} \tag{6}$$

such configurations. Starting from each allowed predecessor word, the search runs until it exits the window or repeats an exact local configuration. Thus it completely decides this finite-window helper question; no time cutoff is used for an individual helper.

Source-node occurrences determine which predecessor words are admissible. A pair can share a row only if the helper realizes the required successor for every admitted context and restores all window bits. Proposed simultaneous folds are replayed in the *patched* table, because one fold can alter another fold's helper. Merely finding a matching in a compatibility graph is not enough.

The selected portfolio has 101 input grammars: the two checked diagonal grammars, the scanner grammar, and 98 saved ties. Searches with one, two, three and five predecessor bits found no additional compatible state-saving fold beyond the existing short helper options. The five-bit context case covers six tape cells including the current cell. Its search is exhaustive only for the unchanged helper rows and the specified single-row entry template; it is not exhaustive over new writable controllers or arbitrary tape simulations. Every examined grammar retains effective AX cost 62, hence total cost 196 in that compiler.

5.4 Combining changes from different packets

The successful-MP rewrite returning its consequent in H and the shorter negation routine are logically different transformations. We combined them with the recent diagonal and separate-diagonal AX layouts rather than assuming their reported savings add.

The shorter negation routine changes the AX sharing opportunities: the two main 196-state variants become 197 states, with 68 readers but only five compatible folds instead of six. Their symbolic contracts pass in the supplied experiments. Across the saved tie portfolio, adding the MP rewrite keeps the MP-only constructions at 196 and the combined NEG/MP constructions at 197; the retained scanner-oriented combination costs 198. These are useful checks of composability, but none is a solution to (1).

Four exact combined tables are included as experimental artifacts:

Construction	States	Source sites	Symbolic contracts
Diagonal gate + MP returning in H	196	3,677	4,785
Separate diagonal head + MP returning in H	196	3,650	4,750
First construction + shorter negation	197	3,461	4,497
Second construction + shorter negation	197	3,434	4,462

For all four, exact blank simulation reaches the recorded clean frame after 58,154 transitions, with every tape cell compared. The symbolic verification uses the inherited boxed register routines. It does not certify the new command-to-calculus transport, the scratch-zero precondition of the shorter negation at every reachable use, or a PA-internal consistency theorem.

5.5 Writable scanner synthesis

The return engine's recognition of a long zero run is another potential optimization point. A one-pass read-only counter is only one implementation; in principle, a writable two-way controller could trade time for states. The exploratory synthesis artifacts formulate exact local contracts restoring all touched cells and test a seven-control positive control before attempting six controls. Solver results are reported with their actual scope: a timeout or unknown result is not UNSAT, and UNSAT for a bounded template is not a general lower bound. No newly certified smaller scanner is used in any claimed machine in this report. The final solver receipt states the completed outcome. The seven-control positive instance is satisfiable and independently replayed. The six-control writable template with a 14-step block bound is unsatisfiable; longer or less restricted queries reported unknown. In particular, the fixed one-symbol action and first-fresh-target restrictions are part of that finite template, not a justified normal form for all possible scanners.

5.6 Recorded completed search scope

Family	Cases/proposals	Outcome
Independent head-parameter substitutions, two grammars	93,312	No cost below 62
Direct head orders, two grammars	1,440	No cost below 62
Mixed semantic and association search on new diagonal words	72,629	Reference 62; changed word best 65
Forced nonidentity-head rotations	62,001	Best changed word 66
Mixed semantic/association search on original 201-token emitter	267,965	Best effective cost 62
Randomized global pair recompression	20,000	Best new source DAG 162
Finite-window helper portfolio (five predecessor bits)	101 grammars	No new compatible fold

The direct-family bounds refer to the imported physical allocation template. The heuristic results are observations of the completed runs, not lower bounds.

6 Scanner audit: a dispatcher return is not an instruction endpoint

The 195-state scanner diagnostic is not a certified consistency machine. Its archived unrestricted symbolic check reports 96 failed contracts out of 4,780. However, one of the accompanying informal diagnostics must be

interpreted carefully: a tape mismatch at the *first* return to the dispatcher does not by itself refute a multi-pass primitive implementation.

We checked this directly on the parent. From the standard source seed, site 25 and registers $(2, 0, \ldots, 0)$, source execution reaches site 0 after 43 steps, with registers

$$(3, 0, 0, 0, 0, 0, 1, 0, 1).$$

For the corresponding admissible gap-57 frame, the 196-state parent returns to native dispatcher 84 after 134 and 290 transitions with intermediate tapes. It reaches the specified site-4 successor frame only after 574 transitions, at its third dispatcher return. A checker that rejected the parent at its first return would therefore reject a correct implementation.

The shortened candidate has a different finite prefix and its proposed symbolic certificate still fails. The point of this correction is not to accept that candidate. It is to keep the logical conclusion accurate: failure of a sufficient simulation certificate is not automatically a counterexample to the blank-run consistency property. Nor does a finite prefix prove that a later rejoining is impossible.

An independent blank-run probe of the shortened candidate completed ten million transitions without a halt or a sampled exact cycle. At that cap it was in native state 0, at head position 18,796, with 4,688 one cells and 64,774 dispatcher visits. This is bounded observational evidence only. It establishes neither nonhalting nor (1).

7 What transfers from ZFC and ConciseIPRA

The ZFC branch supplies valuable implementation techniques: closed source reassociation checked by independent control-flow resolvers, contextual state allocation, arbitrary-value primitive contracts, and positive-duration simulation. Its K215 consistency calibration is about ZFC. Importing that machine or relabeling its states does not change the target to PA.

The current ZFC README carefully distinguishes formal endpoints. It reports compiled external Lean and Rocq correctness for K215, while leaving the internal PA endpoint for K215 open; it separately identifies a checked internal PA equivalence for the literal 240-state machine. We preserve those distinct claims instead of treating all forms of formal verification as interchangeable.

ConciseIPRA provides useful primitive-recursive representations and proof transport methods. The PA comparison's own iPRA bridge files distinguish standard-natural-number bounded simulation from conditional internal endpoints. For example, the endpoint file assumes the machine-specific reductions; the negative-translation files assume appropriate source derivations and witness certificates. A small checker for iPRA does not by itself supply a PA consistency checker, and Python source size does not determine the state count of a complete binary TM with startup.

8 Conclusion and the remaining concrete obligations

The comparison has separated repeated presentations, same-size variants, smaller source programs, rejected 195-state diagnostics and actual physical implementations. It has also checked new combinations and enlarged finite search families. The strongest surviving size in these computations remains 196 working states. The requested strict improvement and full internal biconditional remain unproved here.

A successful next construction must supply the exact table and start convention, a justified source or calculus change, blank initialization, a sound simulation on the admitted frames, and the two arithmetized consistency directions. The proofs and receipts in this archive make those obligations inspectable; they do not replace them with a count of source tokens or a long blank run.

A All incoming archives

ID	Archive	Main finding and status
A01	`Below_U208_Research_2026-10-29.zip`	Same two primary U208 graphs; conditional source pruning removes five primitive sites but grows DAG 197 to 200; local and seven-cell context searches. No validated smaller machine.
A02	`ConPA_below208_research_2026-10-29.zip`	Same two primary U208 graphs; 279,936 parameter permutations in seven emitter rows have minimum 197 source nodes; finite-model obstruction and broader local repairs. No sub-208 construction.
A03	`PA_consistency_TM_research_2026-11-13.zip`	Combined negation/MP program compressed to 157 source nodes at 3573 sites; original source represented with 159 nodes and maximum zero run five; explicit residual MP command compiler. Neither compressed source has a validated sub-196 implementation.
A04	`U208_ConPA_2026-10-29.zip`	Primary U208 local history fold, deleting native 211 into 172. Complete binary table, exact startup, 5084 symbolic instruction cases; mathematical local halting-equivalence proof with inherited consistency calibration.
A05	`U208_ConPA_Conditional_Context_2026-10-29.zip`	Two U208 layouts; four conditional predecessor-context cases give 1/3/7-step full-tape simulation. Same two primary graph classes, not additional stackable savings.
A06	`U208_ConPA_Entry_Context_2026-10-29.zip`	Two primary U208 classes presented by deleting native 172 into 211. Same state quotient up to names; exact startup and complete symbolic contracts.
A07	`U208_ConPA_History_2026-10-29.zip`	Two primary U208 classes with bounded predecessor-history proof; same fold as other primary packets.
A08	`conpa_tm_comparison_and_reduction_search.zip`	Two one-row MP changes: omit redundant zero push, or return successful consequent in H. Explicit adapted command compiler; source savings but still 196. Rejected Shift/combined layouts retained.
A09	`conpa_tm_comparison_and_search.zip`	A/D logical-register swap in same physical state count; two-rotation 195 candidate fails 93 primitive contracts. Includes scanner shortcut and other rejected attempts.
A10	`conpa_tm_study_2026-11-13.zip`	H-only Shift plus MP result in H gives 3788 sites,4928 contracts and 57922-step startup at 196 states. Direct-negation specialization has 3556 sites but 197 states. Repaired 157-node original CFG costs 207 physical states.
A11	`pa196_research_2026-11-13.zip`	Exact original 3812-site CFG represented with 157 internal nodes;158-node factorization loses a needed physical helper. Refined local folds and two-row repair screens; HA-strength native core and canonical proof-stack discussion.
A12	`pa208_research_2026-10-29.zip`	Eighteen additional pairwise nonisomorphic helper variants, all still 208; with two primary classes the U208 family checked for startup and primitive contracts has 20 graphs. Also direct 18-primitive negation, emitter and successful-MP simplifications.
A13	`pa_below196_research.zip`	Earlier 23-archive census,15 selected 196 graphs, four rejected 195 constructions; exact equivalent separation-gate spellings and shortest implication-matrix calculation. No validated sub-196 table.
A14	`pa_consistency_comparison_2026_11_13.zip`	Fresh combinations of source rewrites, split quantifier head with Hilbert certificates, countermodels to axiom deletions, and stronger local deletion searches. No sub-196 result.

ID	Archive	Main finding and status
A15	pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal-vacuity 191-macro/3700-site calculus compiled at 196; separate diagonal head 190-macro/3673-site calculus also at 196. Scanner-oriented variants preserve 196. The shortened-scanner 195 diagnostic has 96 reported unrestricted contract failures; its first-return tape mismatch alone is not an instruction-endpoint counterexample. The U208/PA196 hybrid fails blank startup.
A16	pa_consistency_reduction.zip	New exact-CFG rebracketing at 196, plus H-only Shift 3811-site 196 variant. Integrated equality-witness gate deletes one head and 67 sites (3745 total), but best verified table 203. Two 195 physical outputs have counterexamples/failing contracts.
A17	pa_consistency_research.zip	158-node source representation; compiled regrouping loses one reader but needs one extra scanner, hence 196. Exact 4959 contracts/startup. All 720 head permutations symbolically checked and 5760 order/antecedent variants grammar-screened; neither membership head is freely deletable.
A18	pa_consistency_research_2026-10-29.zip	Comparison of primary U208 family and source-zero pruning. Five fewer source primitives do not yield a physical reduction; calibration scopes retained.
A19	pa_consistency_research_2026-11-13(2).zip	Twelve-rotation 196 source layout; diagonal-vacuity realization reduced 199 to 198; halt-at-last-MP command calculus and bounded physical experiments. No sub-196 table.
A20	pa_consistency_research_2026-11-13.zip	Diagonal-vacuity gate $B \to \forall x_B\, B$ reduces 201 to 191 macros and 3812 to 3700 sites, first realized at 199 states. Direct-negation source 3596 realized at 197. Complete start-up/symbolic cases, written calibration, no smaller table.
A21	pa_tm_comparison_and_reductions.zip	Two new rotated 196 reader layouts; equivalent distributed gate 199-macro/3773-site program and 67-rule grammar realized at 198. Countermodels and physical binding audit separate source from machine proof.
A22	pa_tm_extended_research_2026-11-13.zip	Four closing-decrement 196 variants; zero/positive and difference-bound invariant reduces 4959 generic cases to 4954 admitted cases. Generalized symbolic frame executor and wider searches, no sub-196 result.
A23	pa_tm_research_2026-11-13(2).zip	Same-size zero-jump implementations; purported 195 quotient refuted by exact site 26 frame (wrong opcode in patched parent).202-state parity runtime has symbolic contracts but no certified blank startup.
A24	pa_tm_research_2026-11-13.zip	Shorter-path alternative 196 table; upstream 4959 primitive contracts and startup reportedly rerun in Lean 4.35.0; new variant independently symbolic-checked. Written native-to-PA reverse translation, no exported new PA endpoint.
A25	pa_unary_power_cover.zip	Unary power-covering schema mutually interpretable with PA via conventional proof translations; division-free polynomial emitter and eight-case theorem stream. No compiled state count.
A26	u208_comparison_research_2026-10-29.zip	Nine primary outputs=two graphs. Exact canonical-frame rejection of 207 reassociation: wrong source address/register after 538 steps versus correct parent 568. No sub-208 construction.
A27	u208_research_2026-10-29.zip	Compatibility-parent U208 history fold; one/three/seven-step local simulation,5084 symbolic contracts,48140-step startup plus 141-step bridge; second primary graph class.

B Reproduction and evidence map

Artifact	Purpose
`comparison_agent/inventory.json`	Complete literal-table census, source hashes and member provenance; inferred starts are explicitly marked.
`comparison_agent/archive_comparison.json`	Classification of all 27 top-level archives and their main reported outcomes.
`comparison_agent/census.py`	Rebuild the census against a complete pinned ProveIt checkout using `-repo PATH -out PATH`.
`proof_agent/axiom_variants.py`	Exact baseline emitter and admissible head/gate parameters; the unjustified h1 replacement is disabled.
`research/grammar_search.py`	Randomized closed-sequence recompression.
`research/reskin_search.py`	Mixed axiom-parameter and association search with physical helper scoring.
`research/extended_helpers.py`	Exact two-cell writable helper enumeration.
`research/context_helpers.py`	Finite-window helper enumeration using source-derived predecessor-context sets and simultaneous-fold replay.
`comparison_agent/direct_head_permutations.py`	Exhaustive independent head-parameter substitutions in two fixed input grammars.
`proof_agent/scanner_source_witness.py`	Concrete source and parent-machine witness showing intermediate dispatcher returns.
`evidence/`	Input provenance, aggregate results and construction-combination receipts, with proof-status qualifications.
`SHA256SUMS`	SHA-256 inventory of final deliverables and retained inputs.

The selected executable searches use Python's standard library, except that the separate scanner synthesis needs `z3-solver`. Run the search scripts without Python's `-O` option: inherited symbolic engines use assertions. For deterministic ordering in the new heuristic searches use `PYTHONHASHSEED=0`. Exact proposed counts, rather than elapsed time, are the appropriate comparison coordinate. Solver time limits and heuristic time budgets may produce different trajectories on another machine.

C Primary sources

1. Imported PA196 and PA209 snapshot provenance. The imported upstream revisions are `407cdc356f6b586635e71ac92be394255f4bfea4` (PA196) and `69d98241a721ead20b4f0b17df9062950b40fbfe` (PA209).
2. James Chang, Nonhalting and PA-independence for a 196-state Turing machine.
3. Exact internal implication and independence declarations and external nonhalting declaration.
4. PA196–U208 comparison, PA bridge obligations, and their source-search, archive-audit and reverse-certificate subtrees.
5. All 27 incoming archives. Exact names and findings appear in Appendix A; the machine-readable inventory pins their bytes.
6. ZFC research status and formal endpoint distinctions.
7. ConciseIPRA scope, native calculus and proof-checker metric.

Inherited implementations and inputs retain their supplied license and attribution notices. This report makes no claim that upstream authors reviewed or endorsed the new experiments.